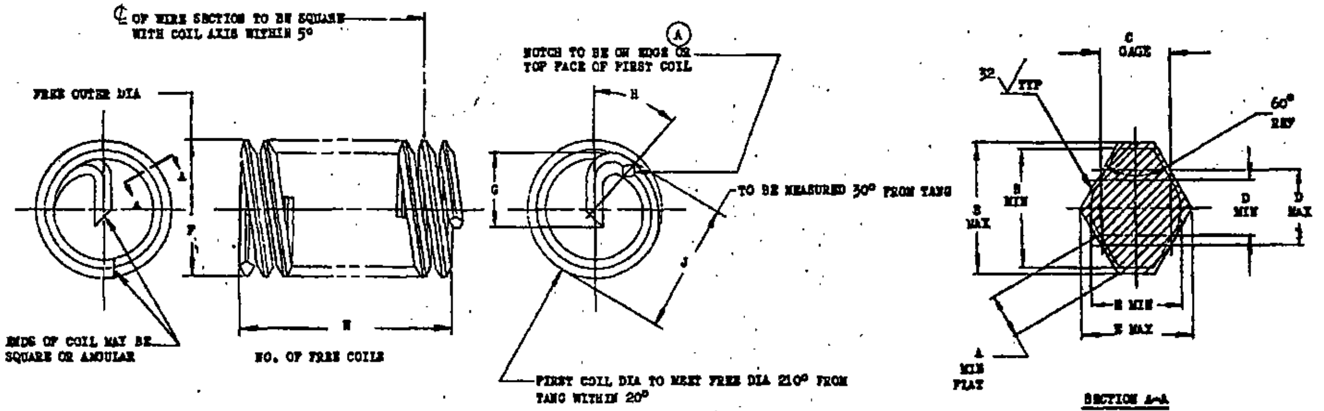
This military standard has been approved by the Department of Defense and is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

Review activities:
Army - AV, GL, MU, MI, WC, EL
Navy - AS
USAF - 11

User activities:
Army - AV
Navy - SH, MC



℄ OF WIRE SECTION TO BE SQUARE WITH COIL AXIS WITHIN 5°
FREE OUTER DIA
A
A
F
ENDS OF COIL MAY BE SQUARE OR ANGULAR
N
NO. OF FREE COILS
Ⓐ
NOTCH TO BE ON EDGE OR TOP FACE OF FIRST COIL
H
G
TO BE MEASURED 30° FROM TANG
J
FIRST COIL DIA TO MEET FREE DIA 210° FROM TANG WITHIN 20°
C GAGE
32 TYP
60° REF
B MIN
B MAX
D MIN
D MAX
E MIN
E MAX
A MIN FLAT
SECTION A-A

1 DIA NOMINAL LENGTH				
COARSE THREAD				
DASH NO.	NOMINAL THREAD SIZE	LENGTH (REF)	APPROX. WEIGHT LBS/100	N ± 1/4 COIL
C0410	.112 (NO.4)-40	.112	.0121	2-3/4
C0610	.138 (NO.6)-32	.138	.0243	2-3/4
C0810	.164 (NO.8)-32	.164	.0355	3-1/2
C1-10	(a).190 (NO.10)-24	.190	.0573	2-7/8
C4-10	(a).250-20	.250	.145	3-3/8
C5-10	(a).3125-18	.312	.232	4
C6-10	(a).375-16	.375	.391	4-3/8
C7-10	(a).4375-14	.438	.667	4-1/2
C8-10	(a).500-13	.500	.794	4-7/8
C9-10	(a).5625-12	.5625	1.235	5-1/8
C1010	(a).625-11	.625	1.515	5-1/4
C1210	(a).750-10	.750	2.778	5-7/8
C1410	(a).875-9	.875	4.545	6-1/4
C1610	(a)1.000-8	1.000	5.882	6-3/8
C1810	(a)1.125-7	1.125	8.392	6-1/8
C2010	(a)1.250-7	1.250	10.437	7
C2210	(a)1.375-6	1.375	14.505	6-1/2
C2410	(a)1.500-6	1.500	17.561	7-1/4

(a) NOT RECOMMENDED FOR NEW DESIGN, SEE MIL-S-7742 FOR RECOMMENDED SERIES.

1 DIA NOMINAL LENGTH				
FINE THREAD				
DASH NO.	NOMINAL THREAD SIZE	LENGTH (REF)	APPROX. WEIGHT LBS/100	N ± 1/4 COIL
F1-10	.190 (NO.10)-32	.190	.0474	4-1/8
F4-10	.250-28	.250	.109	5
F5-10	.3125-24	.312	.180	5-1/2
F6-10	.375-24	.375	.299	6-7/8
F7-10	.4375-20	.438	.459	6-5/8
F8-10	.500-20	.500	.595	7-7/8
F9-10	.5625-18	.562	.826	8
F1010	.625-18	.625	1.010	9
F1210	.750-16	.750	1.695	9-3/4
F1410	.875-14	.875	2.564	9-7/8
F1610	1.000-12	1.000	3.875	9-5/8
F1810	1.125-12	1.125	4.917	11-1/8
F2010	1.250-12	1.250	6.254	12-1/2
F2210	1.375-12	1.375	7.359	13-3/4
F2410	1.500-12	1.500.	8.799	15-1/4

Ⓑ INACTIVE FOR DESIGN AFTER 15 OCTOBER 1969. USE MS122076 THRU MS122275 AND MS124651 THRU MS124850.

Ⓑ INTERCHANGEABILITY RELATIONSHIP: MS122076 THRU MS122275 AND MS124651 THRU MS124850 SUPERSEDE MS21208 PARTS OF THE SAME THREAD SIZE AND NOMINAL LENGTH. THE SUPERSEDED PARTS AND THE SUPERSEDING PARTS ARE COMPLETELY INTERCHANGEABLE AND SHOULD BE STOCKED TOGETHER UNDER THE SUPERSEDING PART NUMBERS.

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Ⓑ FOR CHANGES SEE SHEET 1.

APPROVED 17 Jul 61 REVISED Ⓐ 12 Jan 66 Ⓑ 15 Oct 69

P.A. USAF - 11
AV
Other Cust
Army - AV
Navy - AS

TITLE
INSERT, SCREW THREAD, COARSE AND FINE, FREE RUNNING, HELICAL COIL, CRES

PROCUREMENT SPECIFICATION
MIL-I-8846

SUPERSEDES:

MILITARY STANDARD
MS21208

SHEET 1 OF 3

DD FORM 1 SEP 57 672-1 (Coordinated)

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FED. SUP CLASS
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Review activities:
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Navy - AS
USAF - 11

User activities:
Army - AV
Navy - SH, MC

1-1/2 DIA NOMINAL LENGTH
COARSE THREAD

DASH NO.	NOMINAL THREAD SIZE	LENGTH (REF)	APPROX WEIGHT LBS/100	N ±1/4 COIL
C0415	.112(NO.4)-40	.168	.0196	4-3/4
C0615	.138(NO.6)-32	.207	.0384	4-3/4
C0815	.164(NO.8)-32	.246	.0556	6
C1-15	.190(NO.10)-24	.285	.0899	5
C4-15	.250-20	.375	.196	5-3/4
C5-15	.3125-18	.469	.369	6-5/8
C6-15	.375-16	.562	.625	7-1/4
C7-15	.4375-14	.656	.943	7-3/8
C8-15	.500-13	.750	1.316	7-7/8
C9-15	.5625-12	.844	1.961	8-1/4
C1015	.625-11	.938	2.439	8-1/2
C1215	.750-10	1.125	4.000	9-3/8
C1415	.875-9	1.312	6.250	10
C1615	1.000-8	1.500	9.091	10-1/8
C1815	1.125-7	1.688	12.928	9-7/8
C2015	1.250-7	1.875	16.070	11-1/4
C2215	1.375-6	2.062	22.264	10-1/2
C2415	1.500-6	2.250	26.915	11-1/2

1-1/2 DIA NOMINAL LENGTH
FINE THREAD

DASH NO.	NOMINAL THREAD SIZE	LENGTH (REF)	APPROX WEIGHT LBS/100	N ±1/4 COIL
F1-15	.190(NO.10)-32	.285	.0784	6-7/8
F4-15	.250-28	.375	.155	8-1/4
F5-15	.3125-24	.469	.384	8-7/8
F6-15	.375-24	.562	.420	11
F7-15	.4375-20	.656	.699	10-5/8
F8-15	.500-20	.750	.909	12-3/8
F9-15	.5625-18	.844	1.266	12-1/2
F10-15	.625-18	.938	1.562	14-1/8
F1215	.750-16	1.125	2.554	15-1/8
F1415	.875-14	1.312	4.000	15-1/2
F1615	1.000-12	1.500	6.063	14-7/8
F1815	1.125-12	1.688	7.572	17-1/4
F2015	1.250-12	1.875	9.502	19-3/8
F2215	1.375-12	2.062	11.276	21-3/8
F2415	1.500-12	2.250	13.650	23-1/2

2 DIA NOMINAL LENGTH
COARSE THREAD

C0420	.112(NO.4)-40	.224	.0272	6-3/4
C0620	.138(NO.6)-32	.276	.0525	6-7/8
C0820	.164(NO.8)-32	.328	.0776	8-3/8
C1-20	.190(NO.10)-24	.380	.1330	7-1/8
C4-20	.250-20	.500	.308	8
C5-20	.3125-18	.625	.510	9-1/4
C6-20	.375-16	.750	.862	10
C7-20	.4375-14	.875	1.250	10-1/4
C8-20	.500-13	1.000	1.852	11
C9-20	.5625-12	1.125	2.632	11-1/2
C1020	.625-11	1.250	3.448	11-3/4
C1220	.750-10	1.500	5.263	13
C1420	.875-9	1.750	8.333	13-3/4
C1620	1.000-8	2.000	12.500	14
C1820	1.125-7	2.250	17.464	13-5/8
C2020	1.250-7	2.500	21.653	15-3/8
C2220	1.375-6	2.750	30.122	14-3/8
C2420	1.500-6	3.000	36.268	15-7/8

2 DIA NOMINAL LENGTH
FINE THREAD

F1-20	.190(NO.10)-32	.380	.112	9-1/2
F4-20	.250-28	.500	.212	11-3/8
F5-20	.3125-24	.625	.397	12-1/4
F6-20	.375-24	.750	.553	15
F7-20	.4375-20	.875	.917	14-5/8
F8-20	.500-20	1.000	1.234	16-7/8
F9-20	.5625-18	1.125	1.695	17-1/8
F1020	.625-18	1.250	2.083	19-1/4
F1220	.750-16	1.500	3.448	20-5/8
F1420	.875-14	1.750	5.263	21-1/8
F1620	1.000-12	2.000	8.188	20-7/8
F1820	1.125-12	2.250	10.227	23-3/8
F2020	1.250-12	2.500	12.750	26-1/4
F2220	1.375-12	2.750	15.194	28-7/8
F2420	1.500-12	3.000	18.591	31-5/8

APPROVED 17 Jul 61 REVISED (B) FOR CHANGES SEE SHEET 1.

P.A. USAF - 11
Other Cust
Army - AV
Navy - AS

TITLE
INSERT, SCREW THREAD, COARSE AND FINE, FREE RUNNING, HELICAL COIL, CRES

MILITARY STANDARD

MS21208

PROCUREMENT SPECIFICATION
MIL-I-8846

SUPERSEDES:

SHEET 2

DD FORM 1 SEP 57 672-1 (Coordinated)

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INSERT DIMENSIONS

COARSE THREAD

NOMINAL THREAD SIZE	A MIN	B		C	D		E		F		G		H	J	
		MIN	MAX		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		MIN	MAX
.112(NO.4)-40	.0068	.0241	.0271	.01250	.01594	.01624	.0189	.0219	.144	.159	.070	.105	25° - 100°	.144	.159
.138(NO.6)-32	.0076	.0295	.0338	.01563	.01985	.02030	.0233	.0273	.178	.193	.091	.135		.178	.193
.164(NO.8)-32									.205	.220	.117	.152	20° - 100°	.205	.220
.190(NO.10)-24	.0120	.0410	.0451	.02083	.02656	.02706	.0305	.0365	.244	.259	.139	.181	20° - 90°	.244	.259
.250-20	.0164	.0495	.0541	.02500	.03198	.03248	.0378	.0438	.310	.330	.187	.238	20° - 80°	.310	.330
.3125-18	.0176	.0555	.0601	.02778	.03558	.03608	.0416	.0486	.380	.400	.227	.291		.445	.482
.375-16	.0215	.0629	.0677	.03125	.04009	.04059	.0477	.0547	.452	.472	.271	.354			
.4375-14	.0267	.0724	.0773	.03571	.04589	.04639	.0545	.0625	.526	.551	.318	.416		.518	.536
.500-13	.0273	.0747	.0833	.03846	.04946	.04996	.0591	.0671	.597	.622	.349	.463		.586	.606
.5625-12	.0334	.0852	.0902	.04167	.05363	.05413	.0649	.0729	.669	.694	.401	.521		.656	.676
.625-11	.0351	.0895	.0984	.04545	.05855	.05905	.0715	.0795	.742	.767	.428	.572		.727	.747
.750-10	.0402	.0992	.1083	.05000	.06445	.06495	.0795	.0875	.881	.906	.511	.635	160° - 220°	.823	.843
.875-9	.0466	.1110	.1201	.05556	.07167	.07217	.0892	.0972	1.022	1.052	.578	.714		.967	.992
1.000-8	.0544	.1259	.1351	.06250	.08069	.08119	.1004	.1094	1.166	1.196	.671	.781		1.091	1.116
1.125-7	.0645	.1451	.1546	.07143	.09229	.09279	.1160	.1250	1.315	1.355	.719	.843	165° - 210°	1.249	1.274
1.250-7									1.443	1.483	.781	.937		1.374	1.399
1.375-6	.0775	.1695	.1804	.08333	.10775	.10825	.1368	.1458	1.598	1.641	.922	1.091		1.551	1.576
1.500-6									1.727	1.772	.969	1.187		1.676	1.701

FINE THREAD

NOMINAL THREAD SIZE	A MIN	B		C	D		E		F		G		H	J	
		MIN	MAX		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		MIN	MAX
.190(NO.10)-32	.0076	.0295	.0338	.01563	.01985	.02030	.0233	.0273	.236	.256	.131	.182	10° - 90°	.236	.256
.250-28	.0085	.0347	.0387	.01786	.02270	.02320	.0253	.0313	.306	.326	.178	.244	15° - 75°	.306	.326
.3125-24	.0120	.0410	.0451	.02083	.02656	.02706	.0305	.0365	.380	.400	.209	.291		.359	.377
.375-24									.448	.468	.240	.353		.422	.439
.4375-20	.0154	.0495	.0541	.02500	.03198	.03248	.0378	.0438	.524	.549	.287	.416		.494	.514
.500-20									.592	.617	.318	.479		.556	.576
.5625-18	.0176	.0555	.0601	.02778	.03558	.03608	.0416	.0486	.666	.691	.354	.541		.625	.645
.625-18									.733	.758	.386	.588		.688	.708
.750-16	.0215	.0629	.0677	.03125	.04009	.04059	.0477	.0547	.876	.901	.464	.557		.820	.840
.875-14	.0267	.0724	.0773	.03571	.04589	.04639	.0545	.0625	1.021	1.051	.495	.604		.955	.980
1.000-12	.0334	.0852	.0902	.04167	.05363	.05413	.0649	.0729	1.169	1.199	.526	.728		1.094	1.114
1.125-12	.0334	.0852	.0902	.04167	.05363	.05413	.0649	.0729	1.304	1.334	.557	.819	10° - 70°	1.219	1.244
1.250-12									1.439	1.469	.679	.941		1.344	1.369
1.375-12									1.575	1.610	.789	1.001		1.469	1.494
1.500-12									1.710	1.745	.811	1.123		1.594	1.624

NOMINAL LENGTH: NOMINAL LENGTH EQUALS MINIMUM THROUGH HOLE LENGTH WITHOUT COUNTERSINK AND WITH INSERT LOCATED 1/8 TO 1/2 THREAD PITCH BELOW TOP SURFACE.
FREE COILS: THE NUMBER OF FREE COILS OF THE INSERT ARE COUNTED AS FOLLOWS:
UNC THD-90° FROM TANG FOR SIZES UP TO AND INCLUDING .625-11.
180° FROM TANG FOR SIZES .750-10 AND LARGER.
UNF THD-90° FROM TANG FOR ALL SIZES.

MATERIAL: STEEL, CORROSION RESISTANT, *SEE PROCUREMENT SPECIFICATION.*
THREAD: INSERT, WHEN ASSEMBLED IN BASIC TAPPED HOLE AS SPECIFIED ON STANDARD MS33646, SHALL PRODUCE A FINISHED INTERNAL THREAD, MEETING ALL REQUIREMENTS OF HANDBOOK H-28.
TANG LOCATION: THE INSERT TANG LOCATION SHALL CONFORM TO SPECIFICATION AMS 7245C.
SURFACE ROUGHNESS: MIL-STD-10.
DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED.

THE MS PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER, PLUS THE DASH NUMBER.
EXAMPLE OF PART NUMBER.: MS21208C0415 = A FREE RUNNING HELICAL COIL INSERT (MS21208). COARSE THREAD (C), NUMBER 4 THREAD DIA (04), 1-1/2 *DIA NOMINAL LENGTH* (15)

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.
REFERENCED DOCUMENTS SHALL BE *OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.*

APPROVED 17 Jul 61 REVISED (B) FOR CHANGES SEE SHEET 1.

P.A. USAF - 11 Other Cust Army - AV Navy - AS	TITLE INSERT, SCREW THREAD, COARSE AND FINE, FREE RUNNING, HELICAL COIL, CRES	MILITARY STANDARD MS21208
PROCUREMENT SPECIFICATION MIL-I-8846	SUPERSEDES:	SHEET 3

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Review activities:
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User activities:
Army - AT
Navy - SH, MC